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INTELLIGENT PPE DETECTION AND HAZARD RECOGNITION USING COMPUTER VISION AND EFFICIENTDET

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

DSA 02- COMPUTER VISION WITH OPEN CV

to the award of the degree of

**BACHELOR OF TECHNOLOGY IN
ARTIFICIAL INTELLIGENCE & DATA SCIENCE**

Submitted by

V G ABBIRAMY (192524097)

Under the Supervision of

Dr. G A Senthil

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

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Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Intelligent PPE Detection And Hazard Recognition Using Computer Vision And Efficientdet**” has been carried out by **V G ABBIRAMY (192524097)** under the supervision of **Dr. G A Senthil** and is submitted in partial fulfilment of the requirements for the current semester of the **B.Tech Artificial Intelligence & Data Science** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Dr. Narendra Kumar,
Professor
Intelligent Computing
SIMATS Engineering,
SIMATS, Chennai – 602105.

COURSE FACULTY

Dr. G A Senthil
Professor
Cloud Computing
SIMATS Engineering,
SIMATS, Chennai – 602105.

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

DECLARATION

We, **V G ABBIRAMY (192524067)** of **B.Tech Artificial Intelligence & Data Science**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled **Intelligent PPE Detection And Hazard Recognition Using Computer Vision And Efficientdet** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 04/09/2026

Signature of the Students with Names

V G ABBIRAMY (192524097)

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Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Individual Contribution Statement

Student	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
KANMANI K B (192524306)	Dataset collection, preprocessing, annotation, and system design	EfficientDet model setup, dataset preparation, and PPE detection module development	Model training, validation, and performance evaluation	Chapters 1, 2 & 3	50%
ABBIRAMY (1925240697)	Hazard recognition, interface development, and documentation	Hazard recognition logic, alert generation, and system integration	Testing, result analysis, and comparison of detection performance	Chapters 4 & 5, References & Appendices	50%
Total					100%

ABSTRACT

Workplace safety is a major concern in construction, manufacturing, mining, logistics, and other industrial environments where workers are exposed to various physical and operational hazards. Personal Protective Equipment (PPE), such as helmets, safety vests, gloves, goggles, masks, and safety shoes, plays an important role in reducing the risk of workplace injuries. However, conventional PPE monitoring methods generally depend on manual inspection and continuous supervision by safety personnel. Such approaches can be time-consuming, inconsistent, and difficult to apply continuously, particularly in large working environments. Therefore, an intelligent automated system is required to monitor PPE compliance and identify potentially hazardous situations in real time.

This project proposes an Intelligent PPE Detection and Hazard Recognition System using Computer Vision and EfficientDet. The system uses computer vision techniques to process images or video frames and automatically identify workers, PPE items, and selected unsafe conditions. EfficientDet is employed as the primary object detection model because it provides a suitable balance between detection accuracy and computational efficiency. The proposed system involves dataset collection, image preprocessing, annotation, model training, validation, testing, and performance evaluation. The trained model detects PPE categories using bounding boxes and confidence scores and identifies situations where required protective equipment is missing.

The performance of the system is evaluated using standard object-detection metrics such as precision, recall, F1-score, Intersection over Union (IoU), Mean Average Precision (mAP), and inference performance. Different environmental conditions, including variations in lighting, worker position, object size, and partial occlusion, are considered during testing. The detected violations can be presented through a visual interface and can be used to generate warning notifications for safety monitoring.

The proposed system aims to reduce dependence on manual safety inspection, improve the consistency of PPE monitoring, and provide faster identification of safety violations. The project demonstrates how computer vision and efficient deep-learning-based object detection can support intelligent workplace safety systems and provide a foundation for future integration with real-time surveillance and automated industrial safety infrastructure.

Keywords: PPE Detection, Computer Vision, EfficientDet, Hazard Recognition, Object Detection, Workplace Safety

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CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Workplace safety is an important engineering and operational requirement in environments such as construction sites, factories, warehouses, manufacturing plants, mining areas, and other industrial facilities. Workers operating in these environments may be exposed to falling objects, moving machinery, chemical substances, dust, excessive noise, electrical hazards, and other dangerous conditions. Personal Protective Equipment is therefore used to provide an additional layer of protection against workplace hazards.

PPE includes equipment such as safety helmets, reflective safety vests, gloves, protective goggles, masks, and safety footwear. Although PPE can significantly reduce the possibility of injury, its effectiveness depends on whether workers correctly wear the required equipment. Traditional safety monitoring generally involves supervisors observing workers and manually checking PPE compliance. This approach becomes difficult when a large number of workers have to be monitored simultaneously.

The development of Artificial Intelligence and Computer Vision provides an opportunity to automate this process. A computer-vision system can analyse images or video streams and identify objects based on their visual characteristics. Object detection models can locate workers and PPE equipment using bounding boxes and assign appropriate classes to detected objects.

In this project, EfficientDet is used as the core object detection model. The system is designed to detect PPE-related objects and identify potential safety violations. The overall concept combines image processing, deep learning, object detection, and hazard recognition into a unified safety-monitoring framework.

Important aspects considered in the project include:

- Automatic PPE detection.
- Identification of missing or incorrectly worn PPE.
- Recognition of selected workplace hazards.
- Processing of images or video frames.
- Confidence-based detection.

- Performance evaluation using standard metrics.
- Potential real-time monitoring capability.

1.2 Need for the Project

Manual PPE inspection is effective when the number of workers is small and supervision can be performed continuously. However, industrial environments may contain hundreds of workers, multiple work zones, and continuously changing activities. A safety officer cannot always observe every worker at every moment. Consequently, PPE violations may remain unnoticed until an accident or unsafe event occurs.

An automated computer-vision-based system can provide continuous monitoring support. Instead of depending entirely on human observation, cameras can capture the working environment while the trained model analyses the captured frames. When a worker is detected without the required PPE, the system can identify the violation and provide an appropriate warning.

The project is significant because it combines automated detection with hazard recognition, rather than simply identifying PPE objects.

The major needs addressed are:

- Continuous monitoring of workers.
- Reduction of manual inspection effort.
- Faster identification of PPE violations.
- Consistent safety-rule enforcement.
- Monitoring of multiple workers simultaneously.
- Support for safety officers through automated alerts.
- Improved workplace safety awareness.

1.3 Problem Statement

Existing workplace PPE monitoring systems often rely on manual inspection or conventional image-processing approaches that may not provide reliable detection under changing environmental conditions. Variations in illumination, camera angle, worker movement, object size, background complexity, and partial occlusion make automated PPE detection a challenging computer-vision problem.

The engineering problem is therefore to design and implement an intelligent computer-vision-based system capable of detecting PPE equipment and recognizing selected workplace hazards using EfficientDet, while maintaining suitable detection accuracy and computational efficiency.

The system should be capable of processing input images or video frames, detecting relevant objects, determining PPE compliance, and presenting the detected hazards or violations in an understandable manner.

1.4 Project Aim

The main aim of this project is to develop an intelligent PPE detection and hazard recognition system using Computer Vision and EfficientDet to automatically identify PPE compliance and selected workplace safety hazards from images or video streams.

The system aims to provide an efficient, consistent, and scalable approach for supporting workplace safety monitoring.

1.5 Project Objectives

The project objectives are:

1. To study existing computer-vision-based PPE detection and workplace safety monitoring techniques.
2. To prepare and preprocess a suitable PPE and hazard detection dataset.
3. To annotate relevant PPE and hazard categories for supervised object detection.
4. To develop and train an EfficientDet-based object detection model.
5. To detect PPE equipment such as helmets, safety vests, gloves, goggles, or other selected classes.
6. To identify PPE violations and selected hazardous conditions.
7. To evaluate the trained model using precision, recall, F1-score, IoU and mAP.
8. To test the system under different image and environmental conditions.
9. To develop a suitable visual output/interface for displaying detection results.
10. To analyse the strengths, limitations, and practical applicability of the proposed system.

1.6 Scope

The scope of the project covers the development of a computer-vision-based PPE and hazard detection system using EfficientDet. The system focuses primarily on identifying predefined PPE categories and selected hazards represented in the training dataset.

The project includes:

- Dataset collection and preparation.
- Image preprocessing.

- Image annotation.
- Data augmentation where required.
- EfficientDet model training.
- PPE object detection.
- Hazard recognition.
- PPE compliance analysis.
- Performance evaluation.
- Visual presentation of detection results.

The project does not necessarily cover complete industrial safety management, physical control of machinery, or replacement of qualified safety officers. The system is intended to act as an intelligent monitoring and decision-support tool.

Assumptions

The system assumes that:

- Input images/video are of sufficient quality.
- Required PPE classes are represented in the training dataset.
- Camera views provide sufficient visibility of workers.
- The model is trained using appropriately labelled data.

1.7 Expected Engineering Outcome

The expected outcome is a functional prototype of an intelligent PPE and hazard recognition system that accepts images or video frames and automatically identifies predefined PPE objects and safety violations.

The final system should provide:

- Object detection bounding boxes.
- PPE class labels.
- Confidence scores.
- Identification of PPE violations.
- Hazard recognition results.
- Performance evaluation results.
- Visual output for monitoring.

The expected engineering contribution is the integration of EfficientDet, computer vision, PPE compliance analysis, and hazard recognition into a unified workplace safety monitoring system.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review focuses on research and existing technologies related to PPE detection, computer vision, deep-learning-based object detection, workplace safety monitoring, hazard recognition, and EfficientDet.

Relevant research papers, technical documentation, datasets, and existing safety-monitoring systems are studied to understand the current state of the technology. Particular attention is given to the detection accuracy, computational requirements, datasets used, PPE categories detected, and limitations of existing approaches.

The literature review can be organised around the following areas:

- Traditional PPE monitoring techniques.
- Machine-learning-based PPE detection.
- CNN-based object detection.
- YOLO-based safety monitoring.
- Faster R-CNN-based detection.
- SSD-based detection.
- EfficientDet-based detection.
- Computer-vision-based hazard recognition.
- Real-time workplace monitoring.

2.2 Review of Existing Work

2.2.1 Traditional PPE Monitoring

Traditional PPE monitoring primarily depends on safety officers and supervisors who visually inspect workers. Although human supervision can identify complex safety situations, continuous monitoring is difficult in large environments.

2.2.2 Conventional Computer Vision Methods

Earlier automated approaches use image-processing techniques such as colour segmentation, shape analysis, edge detection, and template matching. These methods may work under controlled conditions but generally struggle with changes in lighting, background and object appearance.

2.2.3 Deep Learning for PPE Detection

Deep-learning models have significantly improved object detection because they can learn visual features directly from training data. CNN-based models can identify PPE objects even when their appearance varies across images.

2.2.4 YOLO-Based Approaches

YOLO models perform object detection in a single-stage framework and are widely used where fast inference is important. They can provide real-time detection but their performance depends on the selected model version, hardware and dataset.

2.2.5 Faster R-CNN

Faster R-CNN is a two-stage object detection framework that generally provides strong detection performance. However, its computational requirements can make deployment in resource-constrained or real-time applications more challenging.

2.2.6 EfficientDet

EfficientDet is designed to achieve an effective balance between accuracy and computational efficiency. Its architecture incorporates efficient feature extraction and feature fusion mechanisms, making it suitable for applications where both detection quality and resource utilisation are important.

2.2.7 Hazard Recognition Systems

Hazard recognition extends PPE detection by analysing whether detected objects and workers satisfy predefined safety conditions. For example, if a worker is detected but a required helmet or safety vest is absent, the system can classify the situation as a potential PPE violation.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual Inspection	Human judgement, flexible	Time-consuming, inconsistent	Provides baseline

Image Processing	Simple implementation	Sensitive to lighting/background	Limited applicability
Faster R-CNN	High detection capability	Computationally expensive	Comparison
YOLO	Fast detection	Accuracy varies by model	Alternative
SSD	Lightweight	May have lower accuracy for small objects	Alternative
EfficientDet	Accuracy-efficiency balance	Requires training and tuning	Selected approach

2.4 Research / Engineering Gap

Although significant research has been performed in automated PPE detection, several challenges remain. Many systems concentrate mainly on detecting individual PPE objects and do not provide a broader interpretation of safety compliance. Detection performance can also decrease when PPE objects are small, partially occluded, poorly illuminated, or viewed from different angles.

Another challenge is achieving an appropriate balance between detection accuracy and computational efficiency. Highly accurate models may require significant computational resources, while lightweight models may sacrifice detection performance.

Therefore, the engineering gap addressed by this project is the development of a system that combines:

- PPE object detection.
- PPE compliance analysis.
- Hazard recognition.
- Efficient deep-learning-based detection.
- Practical computational performance.
- Visual safety monitoring.

2.5 Proposed Contribution

The proposed project contributes an integrated PPE detection and hazard recognition framework based on EfficientDet. Instead of treating object detection as the final output, the system uses detected PPE information to determine whether workers satisfy predefined safety requirements.

The major contribution is the integration of:

Input Image/Video → Preprocessing → EfficientDet Detection → PPE Classification →

Compliance Analysis → Hazard Recognition → Alert/Visual Output

This provides a more complete safety-monitoring approach than simply detecting PPE objects.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Safety Officer	Automatic identification of PPE violations
Site Supervisor	Continuous monitoring support
Worker	Accurate and non-intrusive safety monitoring
Organization	Reduction in safety risks
System Administrator	Easy system configuration and maintenance

3.1.2 Functional Requirements

The system should:

1. Accept image/video input.
2. Preprocess the input.
3. Detect workers and PPE.
4. Classify detected objects.
5. Generate bounding boxes.
6. Calculate confidence scores.
7. Identify missing PPE.
8. Recognize predefined hazards.
9. Display detection results.
10. Generate warnings for detected violations.
11. Store or display relevant detection results where applicable.

3.1.3 Non-Functional Requirements

The system should provide:

- Accuracy: Reliable detection of target PPE classes.
- Performance: Reasonable inference speed.
- Reliability: Consistent results across test images.
- Usability: Simple and understandable interface.

- Scalability: Capability to extend PPE/hazard classes.
- Maintainability: Easy model and dataset updates.
- Security: Protection of captured data and system access.

3.2 Constraints & Applicable Standards

Technical Constraints

The model performance depends on dataset quality, hardware capability, image resolution and environmental conditions.

Computational Constraints

Training deep-learning models may require GPU resources, while real-time inference depends on available CPU/GPU capability.

Environmental Constraints

Lighting, shadows, rain, dust, occlusion and camera position can affect detection accuracy.

Safety Constraints

The system should be treated as a monitoring aid rather than the sole mechanism for ensuring workplace safety.

Ethical Constraints

Images of workers may contain identifiable information. Data collection and storage should therefore be handled responsibly.

For standards, use only standards that you actually use or can substantiate in your project. The template specifically asks for the *specific* standard/code and how it is applied.

Example table:

Standard / Code	Requirement	How Applied
Applicable PPE safety standard	Defines PPE requirements	Used as reference for PPE categories
Data/privacy guideline applicable to deployment	Protect captured information	Restrict access and storage
Software/development standard adopted by project	Maintain code quality	Used during implementation

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
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Detection Accuracy	High	%	High
Precision	High	%	High
Recall	High	%	High
mAP	High	%	High
IoU	Suitable threshold	%	High
Inference Speed	Suitable for intended application	FPS	Medium
Detection Confidence	Configurable threshold	%	High
Input Resolution	Dataset/model dependent	pixels	Medium
Number of PPE Classes	Based on dataset	classes	High

3.4 Alternative Solutions & Evaluation

Alternative 1 – YOLO

YOLO provides fast single-stage object detection and is widely used for real-time applications. It is considered because of its speed and relatively simple deployment.

Alternative 2 – Faster R-CNN

Faster R-CNN provides strong object detection capability using a two-stage detection process. However, its computational requirements can be higher.

Alternative 3 – EfficientDet

EfficientDet provides an efficient object detection architecture designed to balance accuracy and computational cost. It is selected as the proposed model for this project.

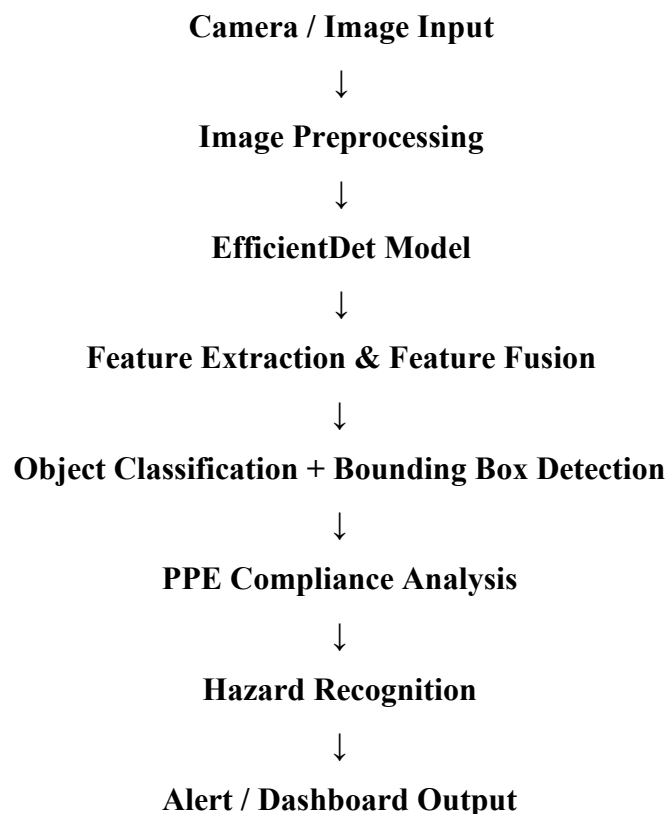
3.5 Selected Design & Detailed Design

Selected Design

EfficientDet is selected as the core detection architecture because the project requires both accurate PPE detection and reasonable computational efficiency. The selected architecture allows the system to process visual information and identify multiple object categories within the same frame.

Proposed System Architecture

The system can be represented as:



Detailed Components

Input Module

The input module receives images or video frames from a camera or stored dataset.

Preprocessing Module

Images are resized and normalized according to model requirements. Data augmentation can be applied during training to improve generalization.

Detection Module

EfficientDet analyses the input image and predicts:

- Object class
- Bounding-box coordinates
- Confidence score

PPE Compliance Module

The detected objects are compared against predefined PPE requirements. For example, if a worker is detected but the required helmet class is not detected within the appropriate region, the system can flag a possible PPE violation.

Hazard Recognition Module

The system interprets the detection results to identify predefined unsafe conditions.

Output Module

The final output displays:

- Bounding boxes
- Object labels
- Confidence scores
- PPE compliance status
- Hazard warnings

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Detection Model	YOLO	Fast	Model-specific limitations	EfficientDet selected based on project requirements
Detection Model	Faster R-CNN	Strong accuracy	Higher computation	Not selected
Input Resolution	High resolution	Better small-object detection	Higher computation	Balanced resolution
Alert Threshold	Low threshold	Higher recall	More false positives	Tuned threshold
Real-time Processing	CPU	Low hardware cost	Lower FPS	GPU preferred where available

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability

The proposed system can reduce dependence on continuous manual monitoring and make better use of existing camera infrastructure. A software-based monitoring solution can also be expanded to additional cameras without requiring a separate physical monitoring device for every location.

Ethics

Worker images may contain personally identifiable information. Therefore, the system should focus on safety-related detection rather than unnecessary identification of individuals. Data should be collected and stored only for legitimate project purposes.

Health & Safety

The system is designed to support workplace safety by identifying missing PPE and predefined hazardous conditions. However, automated predictions should not be treated as a replacement for

professional safety procedures.

Security

Security measures should be considered for:

- Camera feeds.
- Stored images.
- Detection logs.
- User access.
- Model files.
- Dashboard access.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
False detection	Medium	High	High	Improve dataset/training	Medium
Missed PPE detection	Medium	High	High	Data augmentation and threshold tuning	Medium
Poor lighting	High	Medium	High	Diverse training images	Medium
Occlusion	Medium	Medium	Medium	Include occluded examples	Low/Medium
Camera failure	Low	High	Medium	Monitoring/fallback procedure	Low
Data privacy issue	Medium	High	High	Access control and limited storage	Medium
Model performance degradation	Medium	Medium	Medium	Periodic evaluation/retraining	Low/Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows a machine-learning development lifecycle consisting of dataset preparation, preprocessing, annotation, model configuration, training, validation, testing and performance analysis. The dataset is first prepared by collecting images containing workers and PPE-related objects. Images are then annotated according to the required object classes.

After preprocessing and dataset preparation, EfficientDet is trained using the prepared dataset. The model is validated during training to monitor its learning performance and prevent excessive overfitting. After training, the final model is tested using unseen images and relevant performance metrics are calculated.

Resources

- Computer/Laptop
- Python
- Computer Vision libraries
- Deep learning framework
- EfficientDet implementation
- Dataset
- Image annotation tool
- GPU/CPU
- Development environment

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Python	Model development	Implementation
OpenCV	Image/video processing	Preprocessing & testing
EfficientDet	Object detection	Model development
TensorFlow/PyTorch*	Deep-learning implementation	Training
NumPy	Numerical processing	Data processing
Pandas	Dataset handling	Dataset preparation
Matplotlib	Graph generation	Results
Annotation Tool	Bounding-box annotation	Dataset preparation

VS Code / Jupyter	Development	Entire project
GPU	Model training/inference	Training

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
PPE detection	Test labelled images	PPE detected correctly
Worker detection	Test different scenes	Workers identified
Helmet detection	Test helmet/no-helmet images	Correct classification
Vest detection	Test vest/no-vest images	Correct classification
Hazard recognition	Test predefined hazard scenarios	Hazard identified
Confidence	Analyse prediction scores	Above selected threshold
Real-time detection	Video test	Acceptable inference speed
Robustness	Test lighting/occlusion	Reasonable performance

4.4 Results

4.4.1 Training Results

Include:

- Training loss.
- Validation loss.
- Training accuracy where applicable.
- Validation performance.

4.4.2 Detection Results

Show images with:

- Worker bounding boxes.
- Helmet detection.
- Vest detection.
- Other PPE.
- Hazard labels.
- Confidence scores.

4.4.3 Performance Metrics

Include:

Precision

Measures the proportion of predicted positive detections that are actually correct.

Recall

Measures the proportion of actual objects that the model successfully detects.

F1-score

Provides a combined measure of precision and recall.

IoU

Measures the overlap between the predicted bounding box and the ground-truth bounding box.

mAP

Provides an overall measure of object-detection performance across classes.

4.5 Data Analysis & Engineering Judgment

The obtained results should be analysed rather than simply reported. The analysis should determine whether the system meets its intended engineering requirements.

For example, high precision would indicate that most detected PPE objects are correct, while high recall would indicate that the system is able to identify a large proportion of the PPE objects present in the input. If recall is lower than precision, the system may be missing some PPE objects, potentially because of small object sizes, occlusion, poor lighting, or insufficient training examples.

The engineering analysis should also examine false positives and false negatives. False positives can produce unnecessary safety warnings, whereas false negatives are potentially more serious because an actual PPE violation may remain undetected.

Discuss:

- Best-performing PPE class.
- Lowest-performing PPE class.
- Reasons for false positives.
- Reasons for false negatives.
- Effect of lighting.
- Effect of occlusion.
- Effect of object size.
- Effect of camera angle.
- Computational performance.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Study existing PPE systems	Literature review	Fully Achieved
Prepare dataset	Dataset and annotations	Fully Achieved
Train EfficientDet	Training results	Fully Achieved
Detect PPE	Detection outputs	Fully Achieved
Recognize hazards	Hazard outputs	Fully/Partially Achieved
Evaluate model	Metrics	Fully Achieved
Develop visual output	Interface screenshots	Fully Achieved

4.7 Strengths & Limitations

Strengths

The major strength of the proposed system is its ability to automatically analyse visual data and detect PPE-related objects without requiring continuous manual observation. The use of EfficientDet provides an object-detection framework capable of identifying multiple classes within the same image.

Other strengths include:

- Automated monitoring.
- Multiple-object detection.
- Confidence-based predictions.
- Expandable PPE classes.
- Potential real-time application.
- Reduced manual monitoring effort.
- Quantitative performance evaluation.

Limitations

No computer-vision system can guarantee perfect detection under all conditions. The performance of the proposed system depends strongly on the quality and diversity of the training dataset.

Important limitations include:

- Poor lighting can reduce detection accuracy.
- Occlusion can hide PPE objects.
- Small PPE objects can be difficult to detect.

- Unusual camera angles may affect predictions.
- Dataset imbalance may reduce performance for less-represented classes.
- False positives and false negatives may occur.
- Real-time performance depends on hardware.
- The system should not replace professional safety supervision.

4.8 Project Planning, Roles & Budget

Phase	Activity
Phase 1	Problem identification
Phase 2	Literature review
Phase 3	Dataset collection
Phase 4	Dataset annotation
Phase 5	Preprocessing
Phase 6	EfficientDet development
Phase 7	Model training
Phase 8	Testing
Phase 9	Performance evaluation
Phase 10	Interface development
Phase 11	Documentation
Phase 12	Final validation

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project successfully addresses the engineering problem of automated PPE monitoring and hazard recognition using computer vision. The proposed system uses EfficientDet to identify predefined PPE objects and analyse safety-related conditions from images or video frames. The system integrates dataset preparation, image preprocessing, object detection, PPE compliance analysis, hazard recognition and performance evaluation into a unified framework.

The experimental evaluation provides quantitative evidence regarding the effectiveness of the proposed model through precision, recall, F1-score, mAP, IoU and inference performance. The results demonstrate the potential of deep-learning-based computer vision to support workplace safety monitoring and reduce dependence on continuous manual inspection.

However, the system should be considered a safety-support technology rather than a complete replacement for human safety supervision. Its performance is dependent on the training dataset, environmental conditions and computational resources.

5.2 Future Work

Future improvements can focus on making the system more suitable for real-world industrial deployment.

Possible future developments include:

- Real-time CCTV integration.
- Edge-device deployment.
- Mobile application integration.
- Automated SMS/email alerts.
- Cloud-based monitoring.
- Additional PPE classes.
- More advanced hazard categories.
- Multi-camera monitoring.
- Worker tracking.
- Improved small-object detection.
- Better performance under low-light conditions.
- Continuous model retraining.
- Integration with industrial IoT systems.

- Automated safety reports and analytics.

A future version could also combine PPE detection with other computer-vision capabilities, such as fall detection, restricted-zone detection, proximity monitoring around machinery, and unsafe worker behaviour recognition.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

Through this project, the team gains knowledge in:

- Computer vision.
- Deep learning.
- Object detection.
- EfficientDet architecture.
- Dataset preparation.
- Image annotation.
- Model training.
- Performance evaluation.
- Python-based AI development.
- Machine-learning deployment.

Engineering & Professional Skills Developed

The project develops:

- Problem-solving skills.
- Programming skills.
- Dataset management.
- Experimental design.
- Data analysis.
- Technical documentation.
- Team collaboration.
- Project planning.
- Presentation skills.
- Engineering decision-making.

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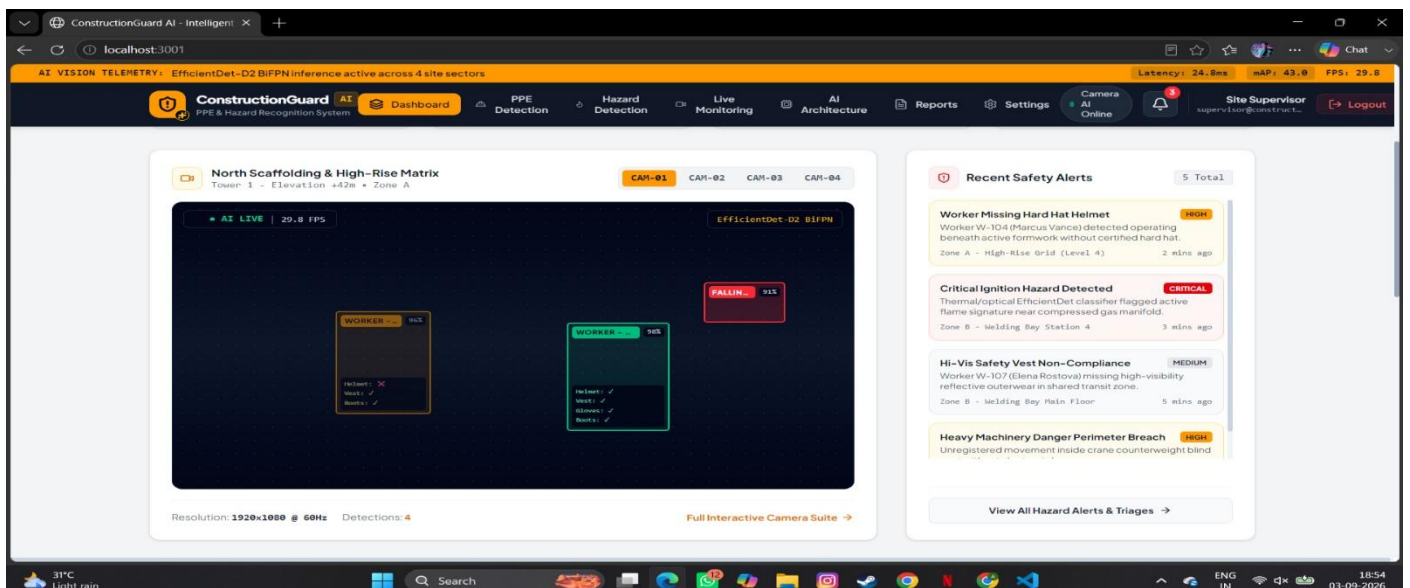
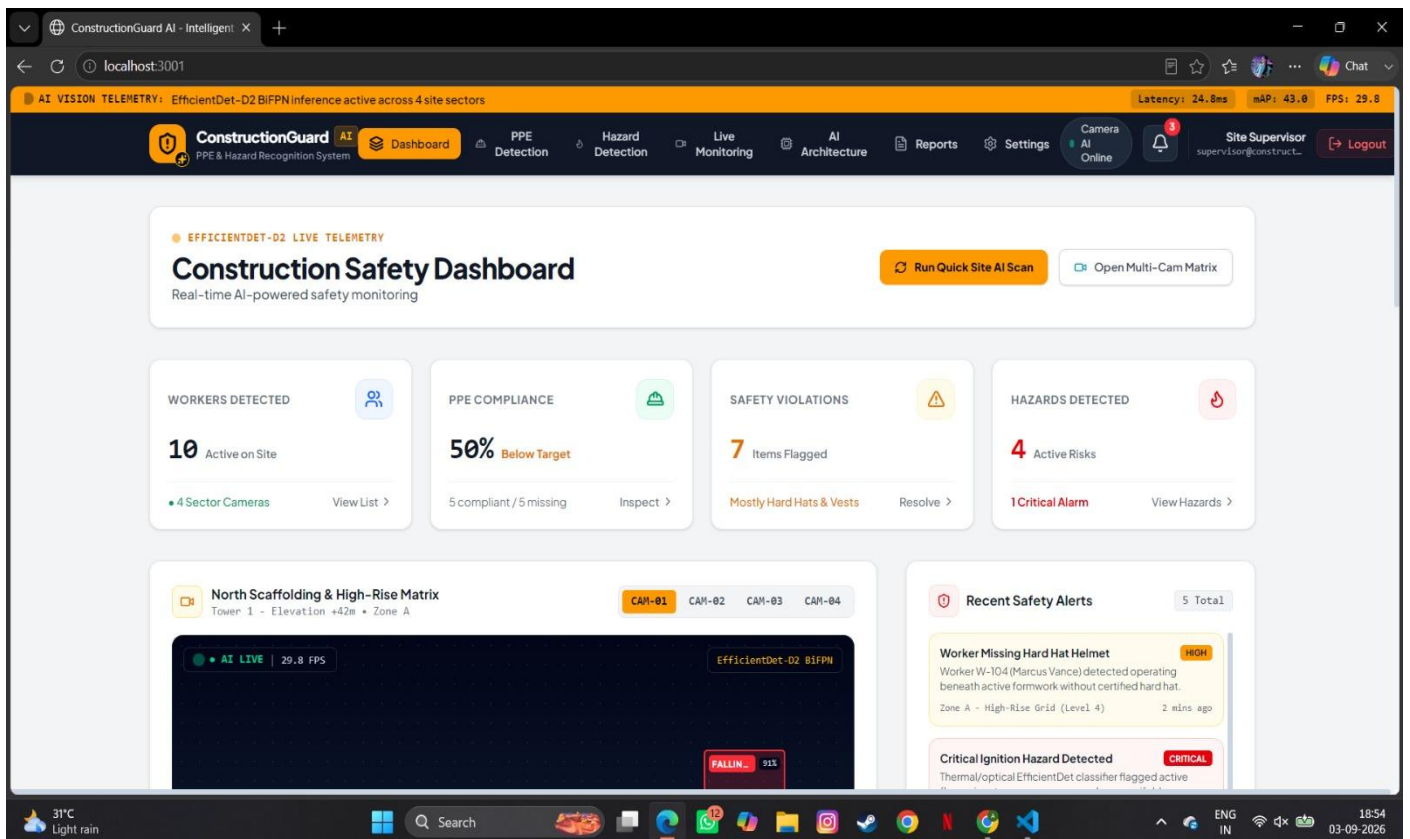
APPENDICES

Appendix A – Detailed Calculations

Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

F1-score = $2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$



ConstructionGuard AI - Intelligent

localhost:3001

AI VISION TELEMETRY: EfficientDet-D2 BIFPN inference active across 4 site sectors

Latency: 24.8msmAP: 43.0FPS: 29.8

ConstructionGuard AI

PPE & Hazard Recognition System

Dashboard

PPE Detection

Hazard Detection

Live Monitoring

AI Architecture

Reports

Settings

Camera AI Online

Site Supervisor
supervisor@construct...

Logout

Critical Focus Area:Hard Hat Enforcement

Live Worker Safety Status

Real-time individual compliance telemetry

View Full 5-Point PPE Matrix →

WORKER ID / NAME	TRADE / ZONE	HELMET	VEST	GLOVES	BOOTS	STATUS	INSPECT
Marcus Vance W-184	Scaffolding Tech Zone A - High-Rise Grid	✗	✓	✓	✓	PPE Missing	Details
Elena Rostova W-187	Steel Fixer Zone B - Welding Bay	✓	✗	✓	✓	PPE Missing	Details
David Chen W-112	Site Engineer Zone C - Central Core	✓	✓	✓	✓	Compliant	Details
Carlos Mendez W-118	Heavy Rig Operator Zone D - Logistics Yard	✓	✓	✗	✓	PPE Missing	Details
Sarah Jenkins W-125	Electrical Foreman Zone A - High-Rise Grid	✓	✓	✓	✓	Compliant	Details

ConstructionGuard AI

DASHBOARD MODULES

PROTOTYPE & RESEARCH

31°C
Light rain

Search

ENG
IN

18:54
03-09-2026

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)